

DASHI Atomic and Periodic-Table Formalism
369 generative constructor, provenance, and verification companion

Canonical source: Docs/papers/drafts/DASHIAtomicPeriodicTable369Formalism.tex
PR: chboishabba/dashi_agda #886
Date: 2026-09-11

CORE GENERATIVE CHAIN

ternary/369 substrate
-> physical realization
-> one-particle representation
-> antisymmetric fermionic occupancy
-> interacting energy / ground-state selector
-> shell occupancy
-> energetic valence quotient
-> shell-completion recurrence
-> periodic-table observable quotient.

Atomic identity is indexed by (Z, N, Ne) ; for a neutral atom $Ne = Z$.
Subshell capacity: $g(l) = 2(2l+1)$.
Principal-shell capacity: $G(n) = 2 n^2$.

The generic constructor is Nat-indexed. Element Z is not a special cell:
 $Z \rightarrow$ admissible fermion state $\rightarrow H_Z \rightarrow \text{Argmin}(H_Z) \rightarrow$ valence class.

Periodicity is recurrence of valence classes across shell-completion events,
not one fixed global translation and not a finite geometric lookup table.

PROVENANCE / PRIOR IMPLEMENTATION RECEIPTS

2025-11-11 dashifine spectral-line atomic tooling (commit 4f1441e...)
2026-02-16 Base369.agda present at dashi_agda repository inception
(init 65266f7..., blob 8dc130c5...): exact 3/6/9 carriers.
2026-02-21 dashifine full-118 periodic-table visualization authored;
2026-03-05 committed as 9f1430ab...; consumes PeriodicTableCSV.csv.
2026-03-06 dashiQ projection/effective-physics programme (47071bed...).
2026-04-30 dashi_agda atom/chemistry recovery carrier + shell filling
strengthening (42e1d740...); explicitly staged, not finished.
2026-06-07 ChemistryFiniteRuleTargets.agda added (062ccb61...): exact
H..Ne occupations and finite Aufbau/Pauli/Hund target surface.
2026-07-07 dashi_lean4 Base369.lean added from zero lines (55132524...);
explicit Lean proof terms for the finite 3/6/9 carrier laws.
2026-07-19 explicit AtomicPeriodicTableRecoveryBoundary (554e8f93...,
PR #101).
2026-09-11 PR #886 consolidates generic constructor + chronology + paper.

IDENTIFIER DISCIPLINE

atom: Wikidata Q9121; periodic table: Q10693;
Pauli exclusion principle: Q131594.
Pauli 1925 primary paper DOI: 10.1007/BF02980631.
Dewey cues: 539.7 atomic/nuclear physics; 546 inorganic chemistry.

QID is semantic identity, Dewey is catalogue classification, DOI identifies a publication object, and a Git commit is an implementation receipt. None of these identifiers alone supplies physical or theorem authority.

VERIFICATION / NON-PROMOTION STATUS

Present and inspected:

- * Base369 Agda source at repository inception.
- * Lean Base369 proof source and its first-addition commit.
- * exact first-ten shell target source in dashi_agda.
- * full-118 dashifine visualization source.
- * PR #886 Agda source, provenance ledgers, validation root and TeX paper.

Historically reported but not reproduced in this PR:

- * MDL/exclusion filling experiment with closure coordinates $Z=2,10,18$;
original executable/script/hash and parameter schedule remain to locate.

Not claimed as freshly mechanically checked in this session:

- * new PR #886 Agda modules (no local Agda executable; focused CI added);
- * dashi_lean4 project build (proof terms inspected, build not rerun);
- * original dashifine full-118 generated GIF artifact (save call located).

Open physical obligations:

- * operator-to-spectrum realization and physical scale setting;
- * ionisation-energy validation;
- * nuclear-stability bridge;
- * bonding / molecular chemistry;
- * empirical recovery across the periodic table.

NON-COLLAPSE LAWS

- triadic cardinality != orbital quantum numbers;
- shell capacity != physical energy ordering;
- full-118 visualization != generative derivation;
- first-ten exact target != full empirical periodic table;
- generic constructor != completed empirical chemistry.

Canonical full manuscript and equations are in the adjacent .tex source.